Conditional Dimensional Explorer

User Manual

Earth Tones Edition

Interactive Conditional Data Visualization Tool

Application type:	Single-page web application (HTML)
Plotting engine:	Plotly.js 3.1.0
Views:	Histogram, Scatter (2D), Scatter3D
Conditioning:	Slider-based, distance-threshold filter
Data input:	CSV upload, synthetic generators
Exports:	CSV

This document explains how to operate the Conditional Dimensional Explorer: how to load or generate data, condition the point cloud, switch between one-, two- and three-dimensional views, and export results.

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1 Introduction

The **Conditional Dimensional Explorer** is a browser-based tool for the interactive, conditional visualization of multivariate data. It transforms every variable into a uniform $[0, 1]$ scale (empirical CDF) and then lets you explore subsets of the data selected by *conditioning* variables. The points closest to a chosen conditioning point are highlighted, while the rest fade toward transparency.

The application supports three visualization modes depending on how many variables you choose to represent:

- **1 variable** — a histogram of the selected variable's original values.
- **2 variables** — a scatter plot in the unit square.
- **3 variables** — an interactive 3D scatter plot.

1.1 Key features

- Load any numeric CSV file (comma, semicolon or tab separated).
- Generate two synthetic datasets: a fixed 15-variable structured sample and a 10-variable realistic correlated sample.
- Condition on any subset of variables using sliders.
- Fade-out by distance: the "distance threshold" slider controls how sharply the cloud fades.
- Save the active dataset as CSV.
- View the data as a table and inspect a Pearson correlation matrix.
- Full-screen mode and an exit button.

No installation required

The application is a single HTML file. It requires an internet connection only on first load, because it fetches Plotly.js from a CDN. All computation runs locally in your browser.

2 How the Visualization Works

2.1 Empirical CDF normalization

Each variable is transformed to a uniform scale on $[0, 1]$ using its **empirical cumulative distribution function** (ECDF):

$$q_{ij} = \frac{\text{rank}(x_{ij})}{n + 1} \tag{1}$$

where n is the number of rows and $\text{rank}(x_{ij})$ is the rank of the j-th column value in row i (with rank 1 the smallest). This makes every axis comparable regardless of units or scale.

2.2 Conditioning and the distance threshold

If you select k conditioning variables $\{c_1, \ldots, c_k\}$, you also get k sliders in the right panel. For each point, a distance in the normalized space is computed:

$$d_i = \sqrt{\sum_{m=1}^{k} \left(q_{ic_m} - \theta_{c_m}\right)^2} \tag{2}$$

where θ_{c_m} is the slider value of the conditioning variable. Points with $d_i < \tau$ (with τ the *distance threshold*) are shown with an opacity

$$\alpha_i = \max\left(0.2,\ 1 - \frac{d_i}{\tau}\right). \tag{3}$$

Points with $d_i \geq \tau$ are hidden. Lowering τ focuses on a small neighbourhood; raising it shows more points.

2.3 The three view modes

Represented	What you see
1 variable	A histogram of that variable's raw (unnormalized) values, filtered by the current conditioning.
2 variables	A 2D scatter plot of their ECDF coordinates $[0,1] \times [0,1]$, with a green frame.
3 variables	A 3D scatter plot of their ECDF coordinates inside a unit cube.

The axes are always ECDF coordinates in 2D/3D

In the 2D and 3D views, the axes show **normalized** values between 0 and 1, so point positions on every axis are comparable. In the histogram view (1 variable), the raw original values are shown on the x-axis.

3 Getting Started

3.1 Opening the application

Open the HTML file in a modern browser (Chrome, Firefox, Edge, Safari). On first load the app automatically generates the fixed 15-variable structured sample, so you can start exploring immediately.

3.2 Interface overview

Region	Content
Top toolbar	Title, file upload, sample-size, data generators, variable selection, save, table, correlation, fullscreen and exit buttons.
Central plot	The active figure (histogram, scatter or 3D).
Right panel	Distance threshold, point counter, and conditioning sliders.
Modals	Variable selection, data table, correlation matrix.

3.3 The point counter

Below the distance-threshold slider you see a badge of the form `2000 / 2000 pts`. The first number is how many points currently satisfy the distance threshold; the second is the total number of rows. This gives immediate feedback on how restrictive your conditioning is.

4 The Toolbar in Detail

Button	Action
Upload CSV	Opens a file picker. Loads a numeric CSV (auto-detects comma, semicolon or tab).
n = (field)	Sample size for the synthetic generators (200–20 000).
Data Initialize	Generates the fixed 15-variable structured dataset.
Realistic ± corr (10 vars)	Generates a 10-variable, correlated sample (Income, Education, Debt, . . .).
Variable selection	Opens the modal to choose represented and conditioning variables.
Save CSV	Downloads the current dataset.
Show data table	Opens a modal with the first 500 rows.
Correlation matrix	Opens a Pearson correlation heatmap.
Fullscreen	Toggles fullscreen.
Exit	Attempts to close the tab (subject to browser rules).

Sample size only affects the generators

The `n =` field controls how many rows the two synthetic generators produce. It has no effect on a CSV you upload; the field updates to reflect the loaded row count.

5 Loading Data

5.1 From a CSV file

Click **Upload CSV** and choose a file. The app:

1. Detects the delimiter (comma, semicolon or tab) from the first non-empty line.
2. Parses the file (supports quoted fields and quoted line breaks).
3. Keeps only rows whose cells are all numeric and complete.
4. If some rows are invalid, shows a confirmation dialog with the first affected row numbers before loading the valid subset.

The first row must be the header (variable names); every subsequent row must have exactly as many numeric values as the header.

Decimal comma

When the file is separated by semicolon or tab, decimal commas are understood (e.g., `3,14`). For comma-separated files the decimal separator must be a point.

5.2 Replacing the active dataset

Whenever you load or generate a new dataset, if a dataset is already active the app asks you whether you want to save it as CSV first. This prevents accidental loss of work.

Irreversible replacement

If you choose not to save, the active dataset, its plots, its modals and the conditioning state are discarded. The new data is installed as a fresh state: two represented variables, one conditioning variable, threshold 0.80 and all sliders at 0.50.

5.3 Synthetic datasets

Two built-in generators are available:

- **Data Initialize** — a fixed 15-variable structure named after academic subjects (math, physics, chemistry, ..., coordination), generated from a five-factor latent model with a fixed loading matrix and a small correlated-factor matrix.
- **Realistic ± corr** — 10 variables (Income, Education, Debt, Savings, Stress, Health, SocialScore, RiskTolerance, WorkHours, LeisureTime) generated from a 10×10 target correlation matrix, adjusted to positive-definite via a small ridge, then drawn through a Cholesky factor.

Both generators clamp every value to $[0, 100]$ and round to one decimal.

6 Choosing Variables

Click **Variable selection** to open the modal. It has two lists:

- **Represented variables** — one, two or three. These are what you actually plot.
- **Conditioning variables** — any number. These become the sliders.

6.1 Rules

- You must select at least 1 represented variable.
- You may select at most 3 represented variables.
- Conditioning variables cannot overlap with represented variables; the app disables those entries in the conditioning list.
- You can choose **None (unconditional)** to remove all conditioning sliders.

Click **Apply & close** to accept. The app rebuilds the sliders and the plot immediately, and then requests fullscreen (a convenience feature that some browsers may block if no user gesture is detected).

Typical workflows

One-dimensional: represent 1 variable and condition on 2–3 others to see how its distribution shifts.
Two-dimensional: represent 2 variables and condition on 1–2 others to see the shape of the cloud from nearby slices.
Three-dimensional: represent 3 variables, no conditioning, to inspect the full cube.

7 Conditioning and the Threshold

Once you have at least one conditioning variable, its slider appears in the right panel. Each slider ranges from 0 to 1 (ECDF units).

7.1 The distance threshold

The top slider of the right panel, labeled *distance threshold*, ranges from 0 to 3. Its value τ determines:

- **Hidden points**: any point with $d_i \geq \tau$.
- **Opacity**: $\alpha_i = \max(0.2,\ 1 - d_i/\tau)$ for the points kept.

Consequently, **smaller** τ means a tighter focus (fewer, brighter points near the conditioning point) and **larger** τ means a looser focus (more points, more uniform alpha).

7.2 The point counter

The badge under the threshold shows `kept / total pts`. Use it to quantify how restrictive the current setting is.

Getting a useful focus

Start with the threshold at 0.80. If the badge shows too many points (e.g., more than half the dataset), lower it; if it shows too few, raise it. When you move a conditioning slider, the badge updates instantly.

8 The Three View Modes

8.1 Histogram (1 represented variable)

The histogram counts the raw, unnormalized values of that variable among the points that pass the conditioning filter. Its x-axis shows the original values, not the ECDF coordinates. This is the fastest way to see how a variable's distribution responds to conditioning.

8.2 Scatter plot (2 represented variables)

Points are shown inside the unit square $[0, 1] \times [0, 1]$ in ECDF coordinates, with a green frame indicating the boundaries. The transparency of each point reflects how close it is to the conditioning point; the axis limits are always $[0, 1]$.

8.3 3D scatter (3 represented variables)

An interactive 3D scatter of the ECDF coordinates inside the unit cube. The Plotly camera controls are enabled (drag to rotate, scroll to zoom). The camera position is preserved while you move the sliders, so you can focus on the region of interest without the view jumping on every update.

Appearance is fixed

In the 2D and 3D views, the axes are always set to $[0, 1]$ so that comparisons across slices remain meaningful. Resetting the view is done through the Plotly modebar's "Home" button.

9 The Data Table and the Correlation Matrix

9.1 Data table

Click **Show data table** to see the first 500 rows. Columns are exactly the header names, and numeric values are rounded to three decimals for readability. If the dataset has more than 500 rows, a footer row states how many are hidden.

9.2 Correlation matrix

Click **Correlation matrix** to compute a Pearson correlation for every pair of variables and render it as a heatmap. The scale is $[-1, 1]$ (blue to red), with the numerical value printed in each cell. This is useful to check whether a conditioning variable is (or is not) correlated with a represented one before selecting it.

10 Saving Your Work

Click **Save CSV** to download the current dataset as a comma-separated file. All variable names are preserved and each value is written verbatim (so re-importing gives the same data). A UTF-8 byte-order mark is prepended so that Excel recognizes the encoding.

Save before generating

Loading a CSV or clicking a generator will prompt you to save the active data if it exists. Always accept the prompt if you want to keep your work.

11 Fullscreen and Exit

Fullscreen toggles the browser's fullscreen mode on the whole page. It is also triggered automatically when you apply a new variable selection.

Exit asks for confirmation and then tries to close the tab. Because of browser security, a page can only close itself if it was opened by a script. If it cannot close, an alert reminds you to close the tab manually.

12 Troubleshooting

"Plotly library could not be loaded"
The page cannot reach the Plotly CDN. Connect to the internet and reload the page. The application cannot plot without Plotly.

"The file must contain at least two variables"
Your CSV has only one column. Make sure the header row contains at least two names.

"No valid numeric rows found"
One or more rows are not fully numeric. If decimal commas are used, make sure the file is semicolon- or tab-separated (not comma-separated).

"Unbalanced quotes"
A quoted field is missing its closing quote. Fix the CSV and try again.

Only a few points are visible
The distance threshold is too small. Raise it, or move the conditioning sliders toward the center (0.5).

The selected variable does not appear as an axis
It is probably in the conditioning list. The app disables overlapping entries in the modal; re-open the modal and adjust the selection.

The full screen does not open automatically
Some browsers block automatic fullscreen if it is triggered outside a user gesture. Use the Fullscreen button instead.

13 Quick Reference

Task	Action
Load a CSV	Upload CSV → pick file.
Generate default data	Data Initialize.
Generate correlated data	Realistic ± corr (10 vars).
Change plotted variables	Variable selection → pick 1, 2 or 3 → Apply.
Change conditioning	Same modal; pick conditioning variables.
Focus tighter	Lower the distance-threshold slider.
Inspect rows	Show data table.
Inspect correlations	Correlation matrix.
Export data	Save CSV.
Fullscreen	Fullscreen button (or automatically after Apply).
Exit	Exit button.

Recommended workflow

1. Load or generate the dataset.
2. Inspect the correlation matrix to understand relationships.
3. Open Variable selection; choose 2 represented variables and 1 conditioning variable as a starting point.
4. Move the conditioning slider to explore different slices.
5. Lower the distance threshold to focus on a local neighbourhood.
6. Switch to 3 represented variables for a full cube view.
7. Save the CSV if you have edited the data.

Version Notes

This manual refers to the version stored as `ConditionalExplorerFINALNEW.html`. Its distinguishing features relative to earlier drafts include:

- Full replacement semantics when loading a new dataset (old state, plots and modals are cleared).
- Safe CSV replacement with an optional save-before-replace prompt.
- Robust CSV parser with delimiter auto-detection and per-row validation.
- Plotly camera preservation across slider updates in the 3D view.
- "None (unconditional)" option in the conditioning list.
- Automatic fullscreen when the user applies a new variable selection.
- Distance-threshold semantics: points with $d_i \geq \tau$ are hidden, opacity decays as $1 - d_i/\tau$.